In [ ]:
```
# time period - 1/1/2015 - 1/1/2026
# takes 5 tickers as input (checks in tickers are valid)
# takes risk free rate as input
# max allocation for 1 stock os 40%
# max sector allocation is 60%
```

In [5]:
```
import yfinance as yf
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import minimize

# 1. User Input with Validation
while True:
    stocks_input = input("Enter EXACTLY 5 stock tickers (comma-separated): ").up
    stocks = [s.strip() for s in stocks_input.split(",") if s.strip()]

    if len(stocks) != 5:
        print("❌ Please enter exactly 5 tickers.\n")
        continue

    try:
        test_data = yf.download(stocks, period="5d",auto_adjust= False)['Adj Clo
        if test_data.empty:
            print("❌ Could not fetch data. Try again.\n")
            continue
        print("✅ Data fetched successfully.")
        break
    except:
        print("❌ Error fetching data. Try again.\n")

print("\nSelected Stocks:", stocks)

# 2. Risk-Free Rate Input
while True:
    try:
        risk_free_rate = float(input("Enter annual risk-free rate (e.g., 0.05 fo
        break
    except:
        print("❌ Enter a valid number.\n")

# 3. Fetch Sector Data (AUTO)
sector_map = {}

for stock in stocks:
    try:
        ticker = yf.Ticker(stock)
        info = ticker.info
        sector_map[stock] = info.get('sector', 'Other')
    except:
        sector_map[stock] = 'Other'

print("\nSector Mapping:")
for stock in stocks:
    print(f"{stock}: {sector_map[stock]}")

sectors = [sector_map[s] for s in stocks]
```

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# 4. Download Historical Data
data = yf.download(stocks, start='2015-01-01', end='2026-01-01',auto_adjust= Fal

returns = data.pct_change(fill_method=None).dropna()
mean_returns = returns.mean()
cov_matrix = returns.cov()

# 5. Portfolio Performance
def portfolio_performance(weights):
    ret = np.sum(weights * mean_returns) * 252
    std = np.sqrt(np.dot(weights.T, np.dot(cov_matrix * 252, weights)))
    return ret, std

# 6. Sharpe Ratio (with RF)
def negative_sharpe(weights):
    ret, std = portfolio_performance(weights)
    return -(ret - risk_free_rate) / std

# 7. Constraints
num_assets = len(stocks)

constraints = [
    {'type': 'eq', 'fun': lambda x: np.sum(x) - 1}  # weights sum to 1
]

# Sector constraint (max 60%)
unique_sectors = list(set(sectors))

for sector in unique_sectors:
    indices = [i for i, s in enumerate(sectors) if s == sector]

    constraints.append({
        'type': 'ineq',
        'fun': lambda x, idx=indices: 0.6 - np.sum(x[idx])
    })

# 8. Bounds (max 40% per stock)
bounds = tuple((0, 0.4) for _ in range(num_assets))

init_guess = num_assets * [1./num_assets]

# 9. Optimization

optimal = minimize(negative_sharpe, init_guess,
                   method='SLSQP',
                   bounds=bounds,
                   constraints=constraints)

weights = optimal.x

# 10. Results
opt_return, opt_std = portfolio_performance(weights)
opt_sharpe = (opt_return - risk_free_rate) / opt_std

print("\n📊 Optimal Portfolio Allocation:")
for stock, w in zip(stocks, weights):
    print(f"{stock}: {w:.2%}")

print(f"\n📈 Expected Return: {opt_return:.2%}")
print(f"📉 Risk (Volatility): {opt_std:.2%}")
```

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print(f"⭐ Sharpe Ratio: {opt_sharpe:.2f}")

# 11. Efficient Frontier
num_portfolios = 5000
results = np.zeros((3, num_portfolios))

for i in range(num_portfolios):
    w = np.random.random(num_assets)
    w /= np.sum(w)

    ret, std = portfolio_performance(w)
    sharpe = (ret - risk_free_rate) / std

    results[0, i] = std
    results[1, i] = ret
    results[2, i] = sharpe

# 12. Plot
plt.figure(figsize=(10,6))
plt.scatter(results[0,:], results[1,:], c=results[2,:], cmap='viridis')
plt.colorbar(label='Sharpe Ratio')

plt.scatter(opt_std, opt_return, color='red', s=100, label='Optimal Portfolio')

plt.xlabel('Risk (Volatility)')
plt.ylabel('Expected Return')
plt.title('Efficient Frontier (Constrained & Sector Balanced)')
plt.legend()
plt.show()
```

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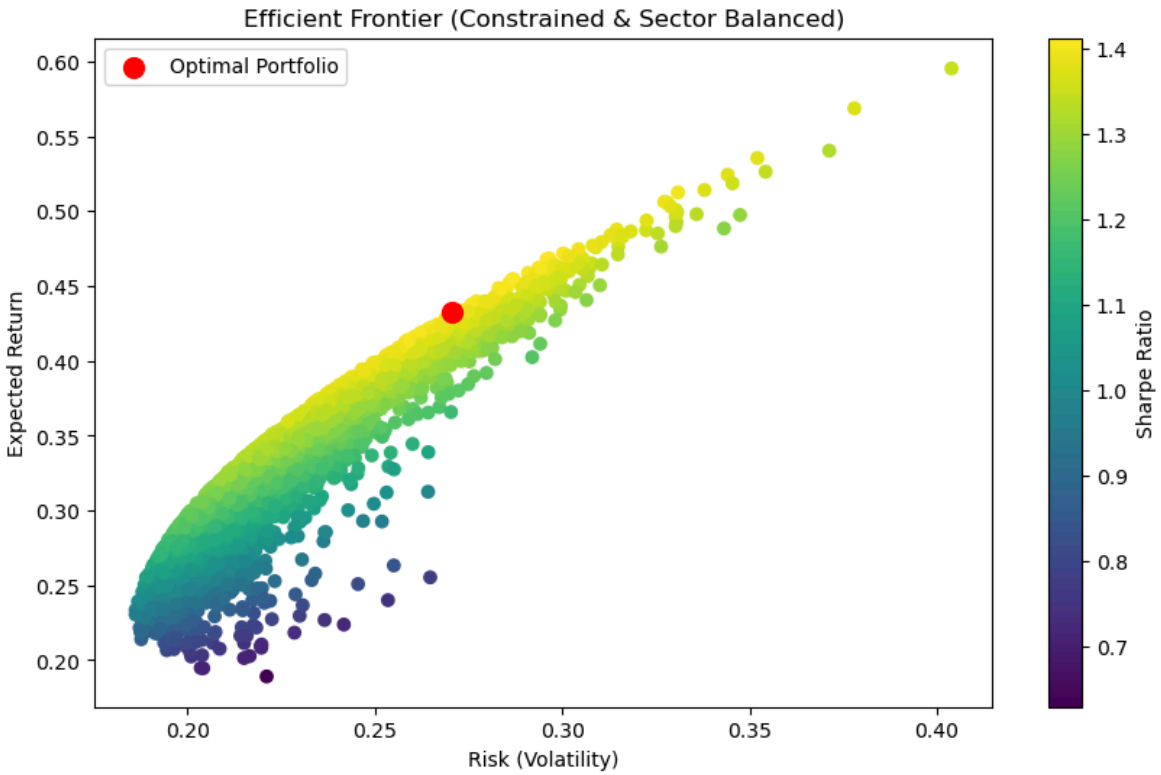
[*********************100%***********************]  5 of 5 completed
✅ Data fetched successfully.

Selected Stocks: ['AAPL', 'GOOGL', 'SBIN.NS', 'NVDA', 'MRF.NS']
Sector Mapping:
AAPL: Technology
GOOGL: Communication Services
SBIN.NS: Financial Services
NVDA: Technology
MRF.NS: Consumer Cyclical
[*********************100%***********************]  5 of 5 completed
📊 Optimal Portfolio Allocation:
AAPL: 0.00%
GOOGL: 27.51%
SBIN.NS: 9.13%
NVDA: 40.00%
MRF.NS: 23.36%

📈 Expected Return: 43.22%
📉 Risk (Volatility): 27.05%
⭐ Sharpe Ratio: 1.41
```


Efficient Frontier (Constrained & Sector Balanced)
Optimal Portfolio
Expected Return
Risk (Volatility)
Sharpe Ratio
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.20
0.25
0.30
0.35
0.40
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7

In [ ]: